Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (Bsc.IT Semester III)

Python for Data Science Practical

Practical - V

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Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c
https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/

5d https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

Input:-

```
import pandas as pd

data = {
  "name": ["Kaif", "Aryan", "Aditya"],
  "age": [19 ,20, 21],
  "qualified": [True, False, False]
}
df = pd.DataFrame(data)

print(df)
```

Output:-

```
      name  age  qualified
0     Kaif   19       True
1    Aryan   20      False
2   Aditya   21      False
```

5b. Write a python code to show statistical information on given data set.

Input:-

1:-

```
# Import statistics Library
import statistics

# Calculate average values
print(statistics.mean([11, 31, 51, 71, 91, 11, 13]))
print(statistics.mean([11, 13, 15, 17, 19, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
```

2:-

```
# Import statistics Library
import statistics

# Calculate average values
print(statistics.median([11, 31, 51, 71, 91, 11, 13]))
print(statistics.median([11, 13, 15, 17, 19, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
```

3:-

```
# Import statistics Library
import statistics

# Calculate average values
print(statistics.mode([11, 31, 51, 71, 91, 11, 13]))
print(statistics.mode([11, 13, 15, 17, 19, 11]))
print(statistics.mode([-11, 5.5, -3.4, 7.1, -9, 22]))
```

4:-

```
# Import statistics Library
import statistics

# Calculate average values
print(statistics.mode([11, 31, 51, 71, 91, 11, 13]))
print(statistics.mode([-11, 5.5, -3.4, 7.1, -9, 22]))
print(statistics.stdev([1, 30, 50, 100]))
```

5:-

```
# Import statistics Library
import statistics

# Calculate the variance from a sample of data
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
```

Output:-

1:-

```
39.857142857142854
14.333333333333334
1.8666666666666667
```

2:-

```
31
14.0
1.05
```

3:-

```
11
11
-11
```

4:-

```
11
-11
41.67633221226007
```

5:-

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
```

5c.Write a python code to create pandas series from dictionaries.

Input:-

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'Kaif': 21, 'Jasad': 20, 'Aryan': 19}

# create a series
series = pd.Series(dictionary)

print(series)
```

Output:-

```
---------------
Kaif      21
Jasad     20
Aryan     19
dtype: int64
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Input:-

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["Kaif", "Jasad", "Aryan", "Aditya"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[60, 70, 65, 70]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
```

2:-

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["Kaif", "Jasad", "Aryan", "Aditya"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[60, 70, 65, 70]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)

# accessing a dataframe using .loc[] function
print(df.loc[True])
```

Output:-

```
          name  degree  score
True      Kaif     MBA     60
False    Jasad     BCA     70
True     Aryan  M.Tech     65
False   Aditya     MBA     70
```

2:-

```
        name  degree  score
True    Kaif     MBA     60
True   Aryan  M.Tech     65
```